

Ahmed Rakin Kamal

PhD Candidate in Theoretical Physics, Masaryk University

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Research Profile

Research interests: higher-derivative corrections and string dualities; string cosmology and moduli stabilisation; time-dependent flux compactifications; generalized symmetries and anomalies; swampland constraints.

Education

2023–present	PhD in Theoretical Physics, Masaryk University — Supervisor: Linus Wulff .
2021–2023	MSc in Theoretical Physics, University of Bologna — Supervisor: Michele Cicoli .
2020–2021	M2 in Mathematical Physics, Université de Bourgogne Franche-Comté — Supervisor: Taro Kimura .
2017–2019	BSc in Physics, BRAC University — Supervisor: Mahbub Alam Majumdar .

Publications

Published

- [1] **Ahmed Rakin Kamal** and Ratul Mahanta. “Time-dependent flux backgrounds in type IIB supergravity”. In: *JHEP* 06 (2026), p. 188. DOI: [10.1007/JHEP06\(2026\)188](https://doi.org/10.1007/JHEP06(2026)188). arXiv: [2512.19793](https://arxiv.org/abs/2512.19793) [[hep-th](#)].
- [2] Dibya Chakraborty, Mishaal Hai, Sayeda Tashnuba Jahan, **Ahmed Rakin Kamal**, and Md Shaikot Jahan Shuvo. “Effect of Moduli Redefinitions on Fibre Inflation”. In: *JCAP* 06 (2026), p. 025. DOI: [10.1088/1475-7516/2026/06/025](https://doi.org/10.1088/1475-7516/2026/06/025). arXiv: [2511.19610](https://arxiv.org/abs/2511.19610) [[hep-th](#)].
- [3] Aviral Aggarwal, Subhrooneel Chakrabarti, Steven Weilong Hsia, **Ahmed Rakin Kamal**, and Linus Wulff. “One loop in $D = 11$ vs $D = 10$: 4-point check”. In: *JHEP* 02 (2026), p. 010. DOI: [10.1007/JHEP02\(2026\)010](https://doi.org/10.1007/JHEP02(2026)010). arXiv: [2506.16391](https://arxiv.org/abs/2506.16391) [[hep-th](#)].
- [4] Steven Weilong Hsia, **Ahmed Rakin Kamal**, and Linus Wulff. “No manifest T duality at order α'^3 ”. In: *Phys. Rev. D* 111.6 (2025), p. L061904. DOI: [10.1103/PhysRevD.111.L061904](https://doi.org/10.1103/PhysRevD.111.L061904). arXiv: [2411.15302](https://arxiv.org/abs/2411.15302) [[hep-th](#)].
- [5] Michele Cicoli, Christopher Hughes, **Ahmed Rakin Kamal**, Francesco Marino, Fernando Quevedo, Mario Ramos-Hamud, and Gonzalo Villa. “Back to the origins of brane–antibrane inflation”. In: *Eur. Phys. J. C* 85.3 (2025), p. 315. DOI: [10.1140/epjc/s10052-025-13982-9](https://doi.org/10.1140/epjc/s10052-025-13982-9). arXiv: [2410.00097](https://arxiv.org/abs/2410.00097) [[hep-th](#)].

Preprints

- [1] **Ahmed Rakin Kamal**. “A Circle That Won’t Return: The Fate of RR Fluxes and D-branes in Type 0A Tachyon Condensation”. In: (June 2026). arXiv: [2606.26280](https://arxiv.org/abs/2606.26280) [[hep-th](#)].
- [2] Dibya Chakraborty and **Ahmed Rakin Kamal**. “Fire at the Tip of the Throat: Hagedorn Phase after brane–antibrane inflation?”. In: (June 2026). arXiv: [2606.08767](https://arxiv.org/abs/2606.08767) [[hep-th](#)].
- [3] Mishaal Hai, **Ahmed Rakin Kamal**, Noshin Ferdous Shamma, and Md Shaikot Jahan Shuvo. “Perturbative Kähler Moduli Inflation”. In: (June 2025). arXiv: [2506.08083](https://arxiv.org/abs/2506.08083) [[hep-th](#)].

Ongoing Research

Doubly Warped Time-Dependent Flux Compactifications and Cosmology. With Ratul Mahanta (USTC), extending our JHEP construction of time-dependent type IIB backgrounds to warped compactifications and studying consistency conditions and cosmological implications.

Non-Reciprocal Gauge Theory. With Nabil Iqbal, I am investigating interplay between non-reciprocal interactions and higher form symmetries.

Research Appointments and Visits

2020–2026	Lecturer (on academic leave), BRAC University, Bangladesh — undergraduate mathematics/physics teaching, research supervision, and digital course development.
May–Jun 2026	Research Visitor, KU Leuven — host: Thomas Van Riet ; work on scale separation and string compactifications; Erasmus+ research mobility grant.
Oct–Nov 2025	Research Visitor, CERN Theory Department — host: Irene Valenzuela ; CALIGOLA mobility grant.
Apr–May 2024	Research Visitor, University of Bologna — collaboration with Michele Cicoli ’s group.
2020	Research Intern, Institut de Mathématiques de Bourgogne, Dijon — worked with Taro Kimura on JT gravity and matrix models.

Selected Talks, Seminars and Conferences

2026	Swamplandia 2026, Catania — lightning talk.
2026	String Phenomenology 2026, Paris — parallel talk: <i>Perturbative Moduli Stabilisation, Inflation and Hagedorn Phase</i> .
2026	String-Pheno Seminars (online) — research seminar: <i>Time-dependent flux backgrounds in type IIB supergravity</i> .

2026	Physics and Mathematics of Generalized Symmetry in QFT, Kavli IPMU — remote participant.
2026	Strings and Geometry 2026, Uppsala — participant.
2025	Integrability, Dualities and Deformations — poster presentation.
2025	KMPB School 2025: D-branes in Physics and Mathematics — participant.
2024	String Phenomenology 2024, Padova — poster presentation.
2024	ICTP String-Math 2024, Trieste — poster presentation.
2024	Accelerated Expansion, dS, and Strings, Madrid — participant.
2024	ESI Landscape vs Swampland, Vienna — participant.
2023	Deconstructing the String Landscape (Landscape), Paris — participant.

Academic Service, Outreach and Organization

- **Co-founder, ICTP Physics Without Frontiers Bangladesh (2024–present):** co-organized QFT and GR courses (approximately 150 students per course), an online research-internship programme, and mentorship activities; delivered a four-lecture Standard Model series.
- **Jamal Nazrul Islam Memorial Winter School, Dhaka:** helped organize the school and secure approximately EUR 20,000 in funding; lecturers included Ashoke Sen, Michele Cicoli, and Suvrat Raju.

Teaching and Research Supervision

2025	ICTP-PWF School on QFT and the Standard Model — four lectures on the Standard Model.
2020–2023	BRAC University — Calculus, Electricity and Magnetism, Linear Algebra, and Complex Analysis.

Research mentoring: Falguni Biswas and Yeshan Juberi (higher-form symmetries/axion physics, 2025–2026); Kazi Fahim Reza Arko and Fahim Hoque (time-dependent backgrounds, 2025–2026); Shavin Shayok Dipro (black-hole information, 2024); Saif Ar Rasul and Sitima Moeen (axion monodromy cosmology, 2024).

Theses

2023	MSc, University of Bologna — <i>Perturbative Moduli Stabilization and String Inflation</i> ; supervisor: Michele Cicoli.
2021	M2, Université de Bourgogne Franche-Comté — <i>Advances in Two-Dimensional Quantum Gravity</i> ; supervisor: Taro Kimura.
2019	BSc, BRAC University — <i>Non-Local Gravitational Interactions and the Black Hole Information Paradox</i> ; supervisor: Mahbub Alam Majumdar.

Awards, Grants and Research Support

2026	Czech Science Foundation (GAČR) research support — project GA26-22343S, <i>One loop in ten and eleven dimensions</i> .
2026	BRAC University research grant (Co-PI) — grant BRACURSGI25014.
2026	Erasmus+ Research Mobility Grant — research visit to KU Leuven.
2025	CALIGOLA Mobility Grant — research visit to CERN.
2021–2023	International Student Merit Scholarship, University of Bologna — ranked first in the relevant merit competition.
2020–2021	MATH4PHYS Merit Scholarship, Université de Bourgogne Franche-Comté
2017–2019	BRAC University — Dean’s List in 6 of 9 semesters; Vice-Chancellor’s Certificate.

Technical Skills and Languages

Scientific computing: Mathematica; Python; LaTeX. **Languages:** Bangla (native); English (fluent).

References

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